MANGALAM COLLEGE OF ENGINEERING
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSD 334 MINI PROJECT
INTERIM EVALUATION 1

QR CODE BASED BILLING SYSTEM IN SUPERMARKET

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ABSTRACT

The objective of this project is to propose a real-time capturing system that helps consumer billing for supermarket shopping using Quick Response (QR) code in smartphones.In supermarkets customers have to take a long queue for billing after shopping and it’s a time-consuming process. Also users will not be aware of the total bill amount at the time of purchase. According to this project, as customers shop they will scan the QR codes of the each items they want to purchase using their smartphone or a scanning device. The scanned items will be added to the shopping cart on their mobile device and the store’s system. At the checkout counter, the customer’s cart is finalized, and the total amount is calculated. A QR code representing the total bill is generated and displayed to the customer. The customer can scan this QR code using their mobile device to complete the payment.

The QR Code-based Billing System will reduce human error, especially in busy supermarket environments.This project will also facilitate quick and secure payments. The front End can be done in HTML, CSS, Javascript or Android and back end in MySQL. Python language can be used. The Reed-Solomon error correction algorithm is used to encode data from the QR code and also it detects and corrects the missing or corrupted data in the code.

INTRODUCTION

- A QR code-based billing system is an advanced digital solution that simplifies shopping and checkout in supermarkets.
- **Speed up the billing process** by allowing the customers to scan QR code
- Ensure **safer, cashless, paperless shopping experience.**
- Customers can **scan products, view details, add or delete item to a digital cart and view total costs** for completing the shopping.
- At payment, customers scan a final payment QR code to complete the purchase.
- **Minimizes long queues** and improves store efficiency.
- Encourages secure and modern payment methods.
- Reduces paper receipts and promotes sustainability.
- Minimize human errors.
- QR code payments **eliminate the need for physical contacts with cashiers**, enhancing hygiene and safety.
- Comfortable environment for customer's shopping.

LITERATURE REVIEW

NO.	Title	Author	Key Concepts	Research Concepts
1.	Smart payment and billing system	Susila sakthy.s, Ragupathy.T, Kishore G	Reduce the time of customers spend at supermarket.User start shopping and add products to virtual cart.Bill generated and payment is made via UPI.	A product is scanned and dropped into the trolley. Now, the cost is revealed on the mobile phone display screen. Another usage of this application is the customer can get rid of the queue by paying the bill by himself through any of the UPI money transfer platforms.

LITERATURE REVIEW

NO.	Title	Author	Key Concepts	Research Concept
2.	Smart Super Market Billing System Using Python	Mrudal Padoli, Apoorva Gupta, Ayush Kumar,bithal Gadekar	Realistic computer vision system for automated billing must be easy and accurate, as a mismatch will result in incorrect billing, which will have a significant financial impact.	Provide guarantee to increase easiness of shopping in mall or other retail shop by scanning QR code of product and payment generating using the python.Computer vision based systems may be built and implemented for automated billing applications that require minimal human intervention and have shorter wait times, resulting in increased customer satisfaction.

LITERATURE REVIEW

NO.	Title	Author	Key Concepts	Research Concept
3.	QR Code Based Secure Billing System For Shops Using Cued Click Points	Mr Vikas Nandganker, Nikita Dongre, Priyanka giri,Diksha Hinge	User scans and add the product to the cart and server act as bill generator send the bill to user. CUED click points are the concept in which persuasive cued click points graphical password schemes are enabled.	System which guarantees to increase the efficiency. Payment security by graphical password using persuasive cued click points.Graphical password will be really helpful to secure the confidential systems. Cued click points is the concept of Persuasive Cued Click point includes usability and security evaluations.

LITERATURE REVIEW

NO.	Title	Author	Key Concepts	Research Concept
4.	Online Super Market System Using QR Scanner	Dr S M Uma, MS T Sindhu, J Aasha, E Muhilan	This web application provide a seamless checkout experience by generating itemized bills based on the products added to the virtual cart. it also helps users to view the cart.	The robust QR codes canning and efficient product expiry date algorithm,the application ensures accurate and timely identification of items,enhancing user trust and safety.
5.	QR Code Based Grocery Billing Expiry Alert System	Dr A R J Sudha, Likith.p	All similar products In that price range as well as discounts or deals.our system uses Qr Code for every product.Also indicates the expiry time of each product.	The model which displays all stock information the system managed via an admin component of this web based application.

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PROBLEM STATEMENT

- Human Errors: Cashiers may make mistakes in product entry or pricing, leading to incorrect billing

- Long Waiting Times: Manual barcode scanning at checkout counters results in congestion and customer frustration.

- Inventory Management Issues: The current system may not update stock levels in real time, leading to discrepancy.

- Limited Checkout Counters: More customers require more billing counters, increasing operational costs.

- Limited Payment Options: Traditional checkout systems may not support seamless digital transactions.

SYSTEM REQUIREMENTS

❖ Frontend Requirements

- Web Development:
  HTML, CSS, JavaScript (React.js / Angular.js / Vue.js)

❖ Backend Requirements

- Programming Languages:
  Python (Django / Flask)
- Frameworks & APIs:
  FastAPI for real-time responses
- QR Code Generation & Processing:
  Python QR Code Library (qrcode)
- Database Requirements
  MySQL / PostgreSQL (Relational)
- Security & Error Handling
  Reed-Solomon Error Correction Algorithm to detect and fix QR code errors
- Additional Software Tools
  IDE (Integrated Development Environment):
  VS Code / PyCharm / Android Studio

PROPOSED METHODOLOGY

System Architecture

- Client Layer (Website): This layer consists of the mobile websites which communicates with the server through APIs.
- Application Layer (Server Side): The server handles requests from the web,processes product and order information, and communicates with external services like payment gateways
- Data Layer (Database & APIs): The database stores product information, user data, and transaction history.

  APIs are used to communicate with external services (like payment gateways) and for the exchange of data between the app and server.
- Payment Gateway Layer: Integration with external payment providers like PayPal, Gpay, Paytm or others to process transactions securely.

PROPOSED METHODOLOGY

Module-Wise Description

- QR Code Generation Module:

  Purpose: To generate unique QR codes for each product

  Description: Each product will have a unique QR code that contains an identifier (URL or product ID).

  Components:

  Product Database: Stores product details.

  QR Code Generator: Converts product ID or URL to a QR code.

  Admin Panel: Allows the admin to update products and re-generate QR codes if necessary.

PROPOSED METHODOLOGY

Module-Wise Description

- QR Code Scanning Module:
  Purpose: To scan QR codes using the user's mobile phone camera.
  Description: When a user scans the QR code, the website decodes it to fetch the product ID or URL. This then sends a request to the backend server to retrieve product details.
  Components:
    Mobile App Camera: Captures the QR code image.
    QR Code Reader: Decodes the QR code into readable data (e.g., URL or product ID)
    API Request: Sends the decoded data to the server to fetch product information.

PROPOSED METHODOLOGY

Module-Wise Description

- Product Information Display Module:
  Purpose: To display product information to the user after scanning the QR code..
  Description: After decoding the QR code,it fetches product details such as name, price and description from the server.
  Components:
    UI Components: Display product details (images, name,price).
    API Request: Retrieves product information from the server.

PROPOSED METHODOLOGY

Module-Wise Description

- Shopping Cart Module
  Purpose: To manage the products added to user's cart.
  Description: Users can add scanned products to their cart,review items,update quantities or remove products.
  Component:
  Cart Database: Temporarily stores the users cart data.
  Ca**art Management**: Handles adding,removing,and updating products in the cart.
  UI Components: Displays the cart contents and total price.

PROPOSED METHODOLOGY

Module-Wise Description

- Checkout and Payment Module:

  Purpose: Handles the checkout process and payment.

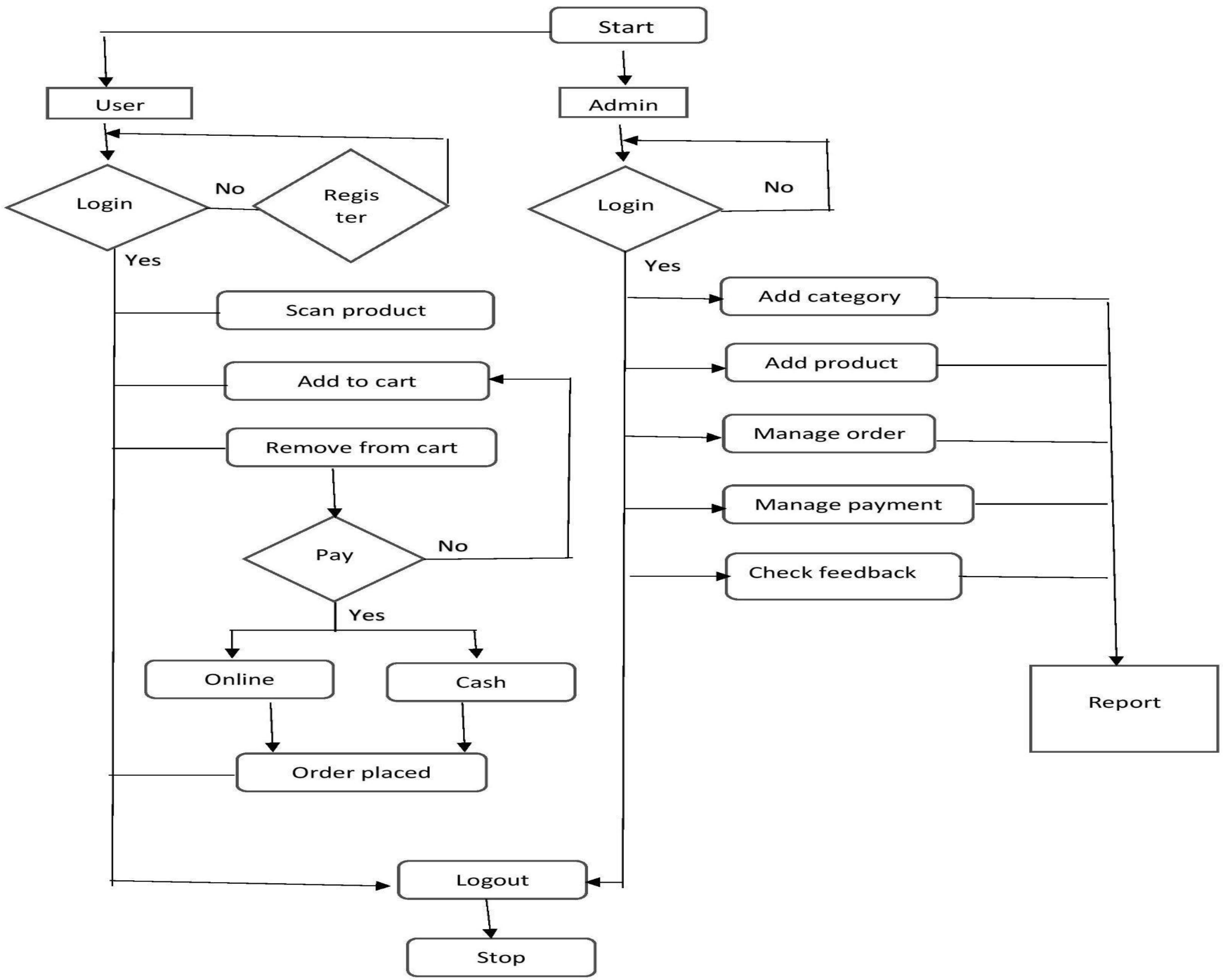
  Description: Users can review final cart before checkout and do payment.

  Components:

  Payment Gateway API: Integrates third-party services like Gpay, PayPal for payment processing.

  Order Database: Stores completed transactions and order history.

DATA FLOW DIAGRAMS



SYSTEM IMPLEMENTATION

1. QR Code Scanning Algorithm
   Implementation: This algorithm is implemented using the library qrcode.js for the web applications.
   Working: When customer reads the qr code, the scanner library processes the image and fetch the details from backend.

2. Product Information Retrieval
   Implementation: When qr code is decoded product ID is passed to backend using API.
   Working: The backend receives product ID, queries the database,fetches the details and send it to the frontend.

1. Cart Management Algorithm
   Implementation: When user adds item, checks if already exist or update the quantity.
   Working: Add items to cart
   Remove items from cart
   Update the cart.

SYSTEM IMPLEMENTATION

4. Payment Gateway Integration

Implementation: Integrates with third party payment apps like Gpay, Paytm

Working: Once the user proceed to checkout, system generates the payment request and processes using this third party apps and then updates the order status and saves the transaction data.

5. Order Management Algorithm

Implementation: After payment an order is created in the database with the required details.

Working: After successful payment, an order is registered with details like the **user ID**, **product IDs**, **order quantity**, **payment status**, and **timestamp**

Also a confirmation message is send to the user..

SYSTEM IMPLEMENTATION

Dataset details

The dataset used for the project includes

1) Product Dataset

   contains information about the products for sale.

   fields like Product ID, Name, Price,Image URL, Stock, Category, etc.

1) User Dataset

   contains details of the users

   fields like User ID, Name, Email, Phno etc

1) Category Dataset

   details about the categories of product

   fields like Category ID, Category name

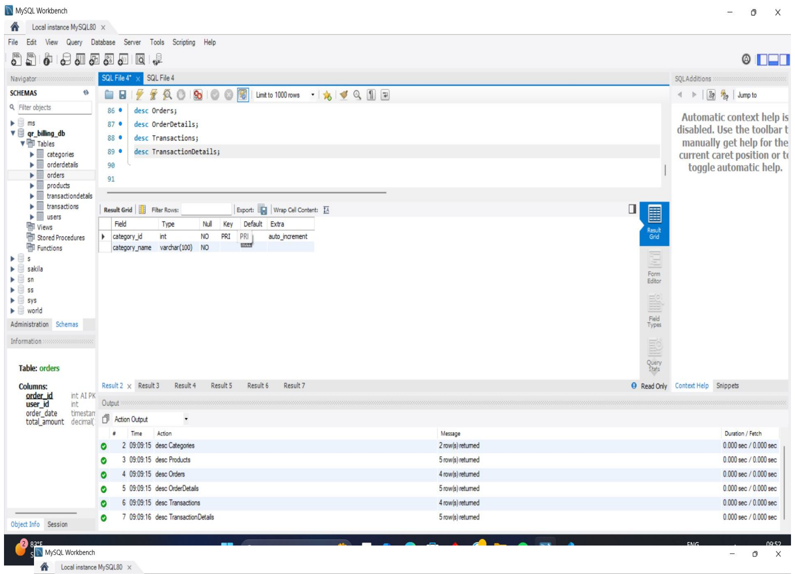
SYSTEM IMPLEMENTATION

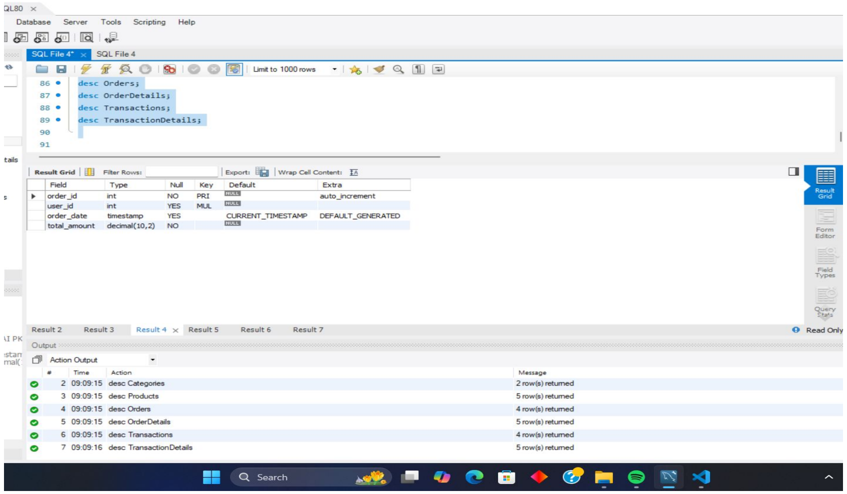
Dataset details

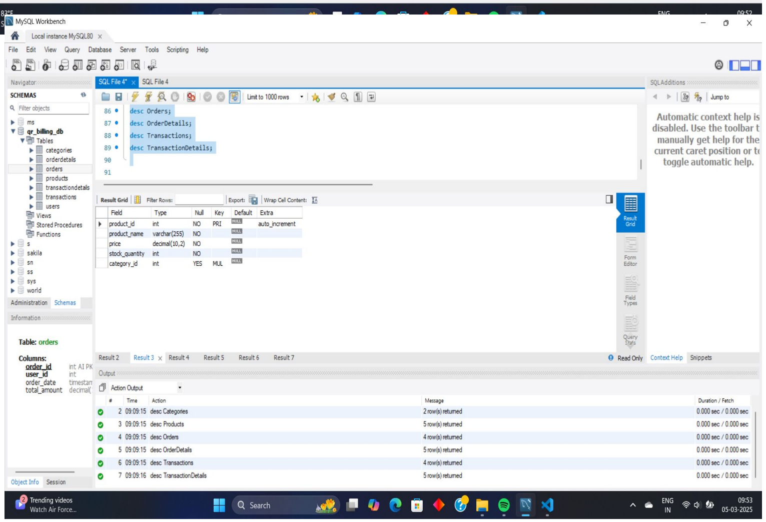
4) Order Dataset
   contains information of completed orders
   entries like Order ID, User ID, Order date and time, Total amount

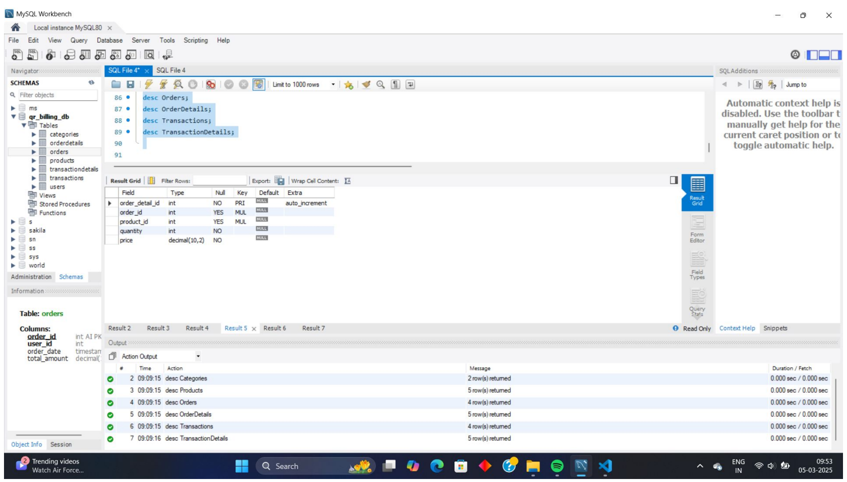
5) Transaction Dataset
   contains the status of transaction
   fields like Transaction ID, Product ID, Time & date of order, total amount etc

RESULTS

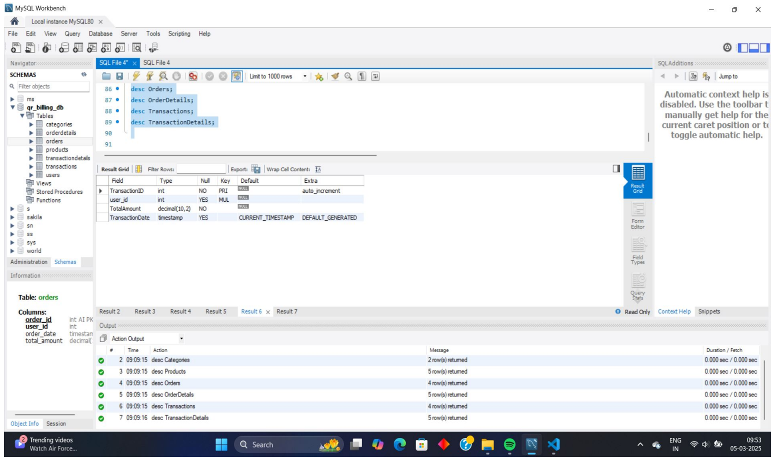


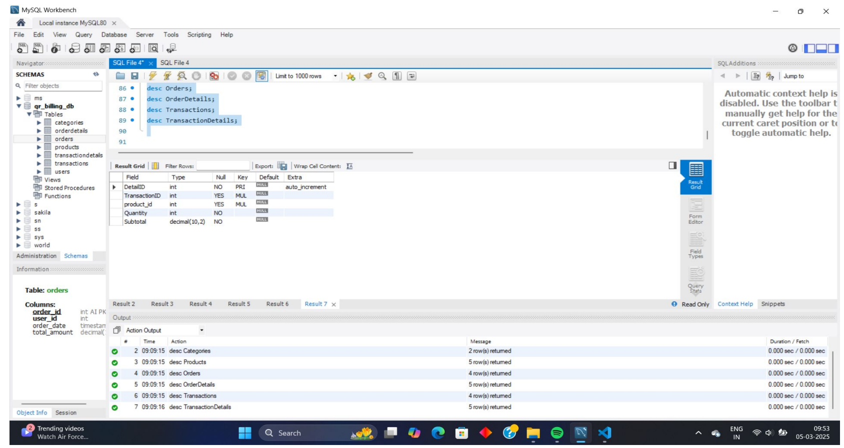

desc Orders;
desc OrderDetails;
desc Transactions;
desc TransactionDetails;
Result Grid
Field Type Null Key Default Extra
order_id int NO PRI auto_increment
user_id int YES MUL
order_date timestamp YES CURRENT_TIMESTAMP DEFAULT_GENERATED
total_amount decimal(10,2) NO
Action Output
2 09:09:15 desc Categories 2 row(s) returned
3 09:09:15 desc Products 5 row(s) returned
4 09:09:15 desc Orders 4 row(s) returned
5 09:09:15 desc OrderDetails 5 row(s) returned
6 09:09:15 desc Transactions 4 row(s) returned
7 09:09:16 desc TransactionDetails 5 row(s) returned





RESULTS


MySQL Workbench
desc Orders;
desc OrderDetails;
desc Transactions;
desc TransactionDetails;
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Table: orders


MySQL Workbench
desc Orders;
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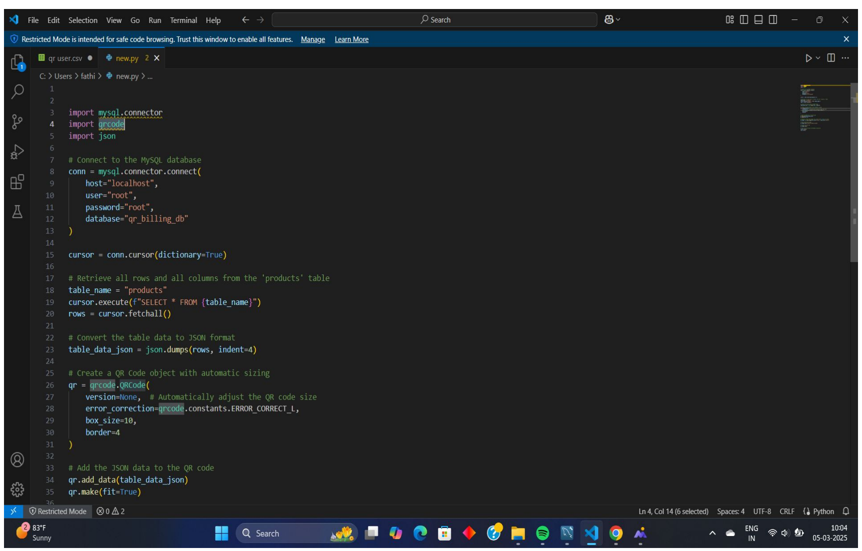
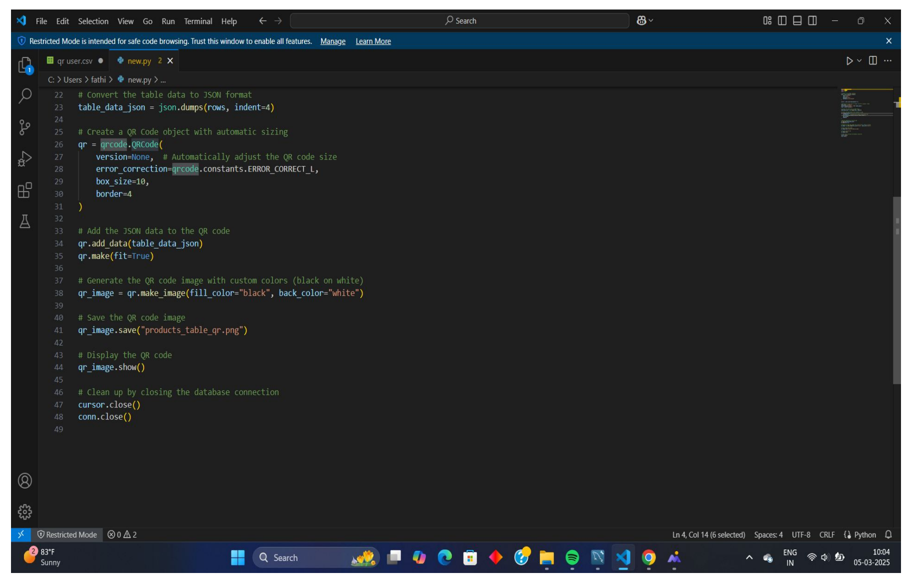

import mysql.connector
import qrcode
import json
# Connect to the MySQL database
conn = mysql.connector.connect(
host="localhost",
user="root",
password="root",
database="qr_billing_db"
)
cursor = conn.cursor(dictionary=True)
# Retrieve all rows and all columns from the 'products' table
table_name = "products"
cursor.execute(f"SELECT * FROM {table_name}")
rows = cursor.fetchall()
# Convert the table data to JSON format
table_data_json = json.dumps(rows, indent=4)
# Create a QR Code object with automatic sizing
qr = qrcode.QRCode(
version=None, # Automatically adjust the QR code size
error_correction=qrcode.constants.ERROR_CORRECT_L,
box_size=10,
border=4
)
# Add the JSON data to the QR code
qr.add_data(table_data_json)
qr.make(fit=True)


# Convert the table data to JSON format
table_data_json = json.dumps(rows, indent=4)
# Create a QR Code object with automatic sizing
qr = qrcode.QRCode(
version=None, # Automatically adjust the QR code size
error_correction=qrcode.constants.ERROR_CORRECT_L,
box_size=10,
border=4
)
# Add the JSON data to the QR code
qr.add_data(table_data_json)
qr.make(fit=True)
# Generate the QR code image with custom colors (black on white)
qr_image = qr.make_image(fill_color="black", back_color="white")
# Save the QR code image
qr_image.save("products_table_qr.png")
# Display the QR code
qr_image.show()
# Clean up by closing the database connection
cursor.close()
conn.close()

CONCLUSION

- The QR Code-Based Billing System significantly reduces checkout time, providing a seamless shopping experience.
  Customers can scan products, manage carts, and make payments without waiting in long billing queues.
- The system can be integrated into existing hypermarket POS systems enhancing efficiency without major infrastructure changes.
- Web dashboard provide an intuitive and smooth user experience for both customers and store administrators.
- A faster and more convenient checkout process improves the overall customer shopping experience, encouraging repeat business.

REFERENCES

- Assist. Prof. Ansar Ahmad, Shakir Khan, Ajay Machhi, Ajay Baria, Akash Sabat ”Store Hub an Android App”: International Journal of computer science and Technology (IJCST)Mar-Apr 2016 International Journal of Computer Science.
- Farley, A., Stevanus,V. (2018) "Swarm Android Mobile Robot: Smart Trolley Application" School of Computer Science Universitas Bina Nusantara.
- Anusha K, Gayana G P, Rekha B Honnali, Sahana G B, and Shridevi H built a "IoT-Based Smart Shopping Cart System" using IOT technology in 2023.
- The group led by James Agajo,Sadiq Thomas, Najashi Gafai, and Ishaya R. Namah from the College of Engineering at Igbinedion University in Okada, Edo State, Nigeria, executed "Development of an Improved Automated Shopping Trolley Payment System Using a Barcode Scanner and Weight Sensor" in December 2021.